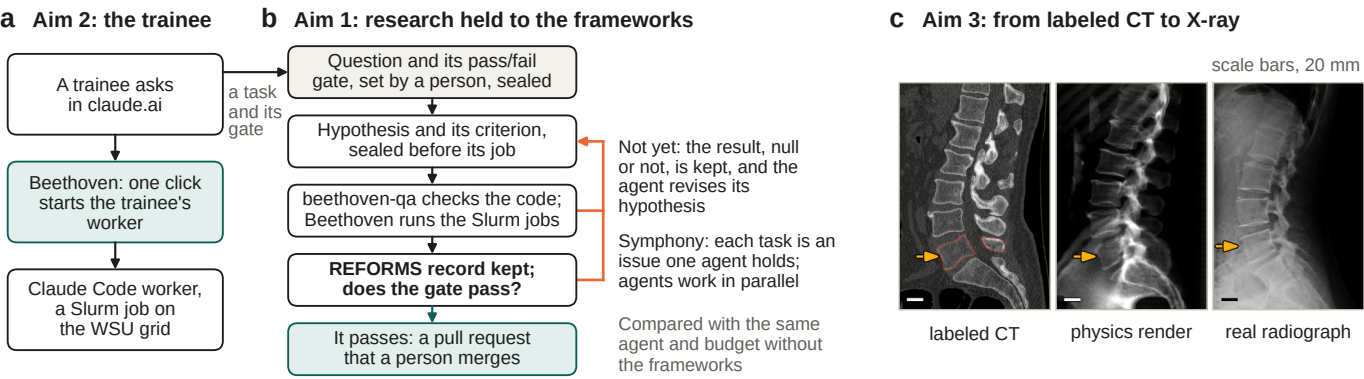

a Aim 2: the trainee
A trainee asks in claude.ai
Beethoven: one click starts the trainee's worker
Claude Code worker, a Slurm job on the WSU grid
a task and its gate
b Aim 1: research held to the frameworks
Question and its pass/fail gate, set by a person, sealed
Hypothesis and its criterion, sealed before its job
beethoven-qa checks the code; Beethoven runs the Slurm jobs
REFORMS record kept; does the gate pass?
It passes: a pull request that a person merges
Not yet: the result, null or not, is kept, and the agent revises its hypothesis
Symphony: each task is an issue one agent holds; agents work in parallel
Compared with the same agent and budget without the frameworks
c Aim 3: from labeled CT to X-ray
scale bars, 20 mm
labeled CT
physics render
real radiograph